LESSON 01 / 09

Sample space first

An experiment has sample space S of possible outcomes; an event is a subset of S. Classical $P(E) = |E|/|S|$ is valid only if the chosen outcomes are equally likely. Two fair dice use 36 equally likely ordered pairs, not 21 equally likely unordered pairs.

INTUITION

Probability is a normalized count

For equally likely finite outcomes, probability is a fraction of the whole sample space. For unequal outcomes, weight them by their probabilities instead of counting them uniformly.

Axioms: $P(S) = 1$, $P(E) \geq 0$, and disjoint events add. Hence $P(\varnothing) = 0$, $0 \leq P(E) \leq 1$, and if $E \subseteq F$, then $P(E) \leq P(F)$.

Complement principle

$$P(E^c) = 1 - P(E).$$

For "at least one", the complement "none" often has fewer cases. For n independent trials, success chance p:

$$P(\text{at least one success}) = 1 - (1-p)^n.$$

WORKED EXAMPLE

Two dice

Sum 7 has six ordered outcomes among 36: probability $1/6$. Writing the ordered space avoids weighting errors: unordered pair $\{1,6\}$ has twice the probability of $\{1,1\}$.

Equally likely versus independent

Equally likely is a property of outcomes in a model; independence is a relation between events. Independent events satisfy $P(A \cap B) = P(A)P(B)$. Neither notion implies the other.

COMMON TRAP

Sample-space checklist

State what is selected, whether order matters, replacement rules, and why outcomes have equal weight. "Favourable over total" without those assumptions is not a probability argument.

LESSON 02 / 09

Overlap correction

For events A, B, disjointly split their union to obtain

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

For disjoint events the intersection is zero; for mutually exclusive events with positive probabilities, they cannot be independent.

WORKED EXAMPLE

Heart or king

In a 52-card deck, 13 hearts and 4 kings overlap at the king of hearts: $P(H \cup K) = (13 + 4 - 1)/52 = 4/13$.

Counting under a probability model

When all selections are equally likely, count using permutations if order matters, combinations if not, then divide. For a committee of size r from N, total outcomes $\binom{N}{r}$. "At least one" often uses a complement count.

WORKED EXAMPLE

Committee has a woman

Choose 3 from 5 men and 4 women. Probability at least one woman is $1 - \binom{5}{3}/\binom{9}{3} = 37/42$. The event and total use the same uniformly selected committee model.

Conditional probability shrinks the space

For $P(B) > 0$:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Multiplication rule: $P(A \cap B) = P(A \mid B)P(B)$. Order matters: $P(A \mid B)$ generally differs from $P(B \mid A)$.

WORKED EXAMPLE

Even die, then six

Given an even result, the reduced space is $\{2,4,6\}$. Exactly one outcome is 6, so $P(6 \mid \text{even}) = 1/3$.

COMMON TRAP

Without replacement

The second draw's probability changes after the first. Use conditional probability; do not multiply the original marginal probabilities as if independent.

LESSON 03 / 09

Independence is factorisation

Events A, B are independent iff $P(A \cap B) = P(A)P(B)$; equivalently, if $P(B) > 0$, $P(A \mid B) = P(A)$. Pairwise independence of three events does not guarantee mutual independence.

COMMON TRAP

Exclusive is not independent

If A, B are disjoint and both have positive probability, $P(A \cap B) = 0$ but $P(A)P(B) > 0$, so they are dependent.

Partition into causes

If $B_1, \ldots, B_n$ are disjoint and cover S, then

$$P(A) = \sum_i P(A \mid B_i)P(B_i).$$

The partition must be exhaustive and its branches disjoint.

Bayes reverses the tree

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_j P(A \mid B_j)P(B_j)}.$$

Numerator = prior × likelihood; denominator = total probability of observed evidence.

WORKED EXAMPLE

Factory defect

Machines A,B,C make 50%,30%,20%, with defect rates 1%,2%,3%. Overall defect chance is $0.5(0.01) + 0.3(0.02) + 0.2(0.03) = 0.017$. Given defective, source A probability is $0.005/0.017 = 5/17$.

COMMON TRAP

Base-rate trap

Sensitivity $P(+ \mid D)$ is not the posterior $P(D \mid +)$. The latter depends on prevalence and false-positive rate through Bayes' denominator.

LESSON 04 / 09

A random variable turns outcomes into numbers

Discrete X has pmf $p(x) = P(X = x)$, with $p(x) \geq 0$ and $\sum_x p(x) = 1$. For finite support:

$$E[X] = \sum_x x p(x), \qquad E[X^2] = \sum_x x^2 p(x)$$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2.$$

Standard deviation $\sigma = \sqrt{\mathrm{Var}(X)}$ has the original units; variance has squared units.

SHORT DERIVATION

Variance as spread about the mean

Expand $E[(X-\mu)^2]$ with $\mu = E[X]$: $E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2$. Variance is nonnegative and is zero exactly when X is constant with probability 1.

Linear changes and indicators

$E[aX + b] = aE[X] + b$; $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$. For indicator I_A, $E[I_A] = P(A)$ and $\mathrm{Var}(I_A) = p(1-p)$.

WORKED EXAMPLE

Three-point distribution

If $X = 1, 2, 3$ with probabilities $0.2, 0.3, 0.5$, then $E[X] = 2.3$, $E[X^2] = 5.9$, variance $= 5.9 - 2.3^2 = 0.61$, SD ≈ 0.781.

COMMON TRAP

Variance of a sum

In general $\mathrm{Var}(X + Y) = \mathrm{Var}\, X + \mathrm{Var}\, Y + 2\mathrm{Cov}(X, Y)$. The two-variance shortcut needs zero covariance (independence suffices); expectation linearity needs no independence.

LESSON 05 / 09

Binomial: fixed independent trials

n independent Bernoulli trials, each success chance p, $q = 1 - p$. Number of successes X has

$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \ldots, n$$

$$E[X] = np, \qquad \operatorname{Var}(X) = npq.$$

The choose factor selects which k trials succeed; each such pattern has probability $p^k q^{n-k}$.

WORKED EXAMPLE

Exactly three successes

For 10 independent trials with $p = 0.2$, $P(X = 3) = \binom{10}{3}(0.2)^3(0.8)^7 \approx 0.201327$. "At least" can be summed or complemented.

COMMON TRAP

Model conditions

Fixed number of trials, two outcomes per trial, constant success probability, and independence. Without-replacement sampling is generally hypergeometric, not binomial.

Geometric waiting time

Let X count trials up to and including first success, $0 < p \le 1$. Then $P(X = k) = q^{k-1}p$ for $k \ge 1$, $E[X] = 1/p$, $\operatorname{Var}(X) = q/p^2$. If instead counting failures before success, support begins at 0: state the convention.

INTUITION

Why the mean is reciprocal

The tail probabilities are $P(X \ge k) = q^{k-1}$. Their sum is $E[X] = 1 + q + q^2 + \cdots = 1/(1 - q) = 1/p$.

A geometric waiting time is memoryless: $P(X > m + n \mid X > m) = P(X > n)$ for nonnegative integers m, n.

LESSON 06 / 09

Data and frequency notation

Observations x_i with frequencies f_i; total $N = \sum f_i$. For ungrouped data, frequencies may all equal 1. Arithmetic mean:

$$\bar{x} = \frac{\sum_i f_i x_i}{N}.$$

It is the balance point and minimizes the sum of squared deviations, not the sum of absolute deviations.

Median and mode

Sort the N observations. For odd N, median is position $(N+1)/2$; for even N, average of positions $N/2$ and $N/2+1$. Mode is the most frequent value; multiple modes are possible.

WORKED EXAMPLE

Five observations

Data $2, 3, 3, 7, 10$: mean $= 25/5 = 5$, median $= 3$, mode $= 3$. Outlier 10 pulls the mean more than the median.

For a discrete frequency table, locate the middle cumulative-frequency positions. For continuous grouped classes of common width h, let l be the lower boundary of the median class, c_f cumulative frequency before it, and f its frequency. Linear interpolation gives $\text{Median} = l + \frac{N/2 - c_f}{f}h$. This estimates a grouped distribution, not the unknowable exact raw median.

Mean deviation

Mean deviation about a reference a is the average absolute distance:

$$MD_a = \frac{\sum_i f_i |x_i - a|}{N}.$$

Often asked about the mean or median; specify the centre. Absolute values prevent positive/negative deviations cancelling.

COMMON TRAP

Use absolute values

$\sum f_i(x_i - \bar{x}) = 0$ by construction, so the signed mean deviation is not a spread measure. Use $|x_i - a|$.

KEY RESULT

Grouped continuous classes

Use class marks $x_i = (\text{lower boundary} + \text{upper boundary})/2$ as representative observations for mean/variance estimates. Results are estimates because original within-class positions are unknown.

LESSON 07 / 09

Population variance and standard deviation

For data x_i with frequencies f_i, mean $\bar{x}$ and $N = \sum f_i$:

$$\sigma^2 = \frac{\sum_i f_i (x_i - \bar{x})^2}{N} = \frac{\sum_i f_i x_i^2}{N} - \bar{x}^2, \qquad \sigma = \sqrt{\sigma^2}.$$

This syllabus convention treats the listed data as the full distribution. Sample variance using divisor $N - 1$ is a different inferential estimator; do not silently substitute it.

SHORT DERIVATION

Computational identity

Expand $\sum f_i (x_i - \bar{x})^2 = \sum f_i x_i^2 - 2\bar{x}\sum f_i x_i + N\bar{x}^2 = \sum f_i x_i^2 - N\bar{x}^2$.

Shift and coding shortcuts

Adding constant c shifts mean and median by c, leaves variance/SD unchanged. Multiplying by a scales mean by a, variance by a^2, SD by $|a|$. For equal class width h, coded values $u_i = (x_i - A)/h$ give $\sigma_x = h\sigma_u$.

WORKED EXAMPLE

Frequency table

Values $1, 2, 3$ with frequencies $1, 2, 1$; $N = 4$, mean 2, mean square $= (1 + 8 + 9)/4 = 4.5$, variance $= 0.5$, SD $= \sqrt{0.5}$.

Grouped continuous data

For classes $[a_i, b_i)$ with frequency f_i, class mark $m_i = (a_i + b_i)/2$ approximates every value in the class. Use m_i in the frequency formulas for mean and variance. Median class is first cumulative frequency at least $N/2$; estimated median is $l + \frac{N/2 - c_f}{f_m} h$. For a unique modal class in equal-width classes of width h, lower boundary l, modal-class frequency f_1 and adjacent frequencies f_0, f_2, the standard interpolation estimate is $\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} h$. These interpolations assume a reasonably smooth distribution within/around the relevant classes.

WORKED EXAMPLE

Grouped table estimate

Classes $[0,10), [10,20), [20,30)$ have frequencies $4, 8, 4$ $(N = 16)$. Midpoints $5, 15, 25$ give mean 15, variance 50, SD $\sqrt{50}$. The middle class has $l = 10, c_f = 4, f_m = 8$, so grouped median estimate is $10 + (8 - 4)10/8 = 15$. Its mode estimate is also 15 by symmetry $(f_0 = f_2 = 4)$. Mean deviation about 15 is $(4 \cdot 10 + 8 \cdot 0 + 4 \cdot 10)/16 = 5$. All are class-based estimates, not exact raw-data statistics.

COMMON TRAP

Unequal class widths

For a histogram with unequal class widths, bar height is frequency density f_i/width_i, not raw frequency. The modal class is determined by density when comparing unequal-width histogram bars.

LESSON 08 / 09

A workflow that prevents most errors

1. Define outcomes / variable / data convention.
2. State equally-likely, independence, replacement, or class-mark assumptions.
3. Choose count, conditional probability, Bayes, distribution, or descriptive statistic.
4. Compute exactly when possible; check range, units, and limiting cases.

WORKED EXAMPLE

Two cards without replacement

Probability both are aces: $(4/52)(3/51) = 1/221$. The second conditional probability is $3/51$, not $4/52$; equivalently $\binom{4}{2}/\binom{52}{2}$.

Useful identities

For independent X_i: $E[\sum X_i] = \sum E[X_i]$ always; $\mathrm{Var}(\sum X_i) = \sum \mathrm{Var}(X_i)$ under independence. For Bernoulli indicators, $X = \sum I_i$ yields binomial mean and variance immediately.

KEY RESULT

Range and sanity checks

Probability lies in $[0, 1]$; variance is nonnegative; SD has original units; mean lies between min and max. A grouped estimate cannot recover exact within-class data.

Official coverage and source gap

JEE Advanced includes probability plus descriptive statistics (mean, median, mode, mean deviation, variance and standard deviation for grouped/ungrouped data, and mean/variance of a random variable). The repository's Probability module covers probability deeply but does not provide a dedicated descriptive-statistics treatment; these statistics sheets are authored to the syllabus and standard definitions, with the class-mark approximation stated.

COMMON TRAP

Not a sampling-statistics chapter

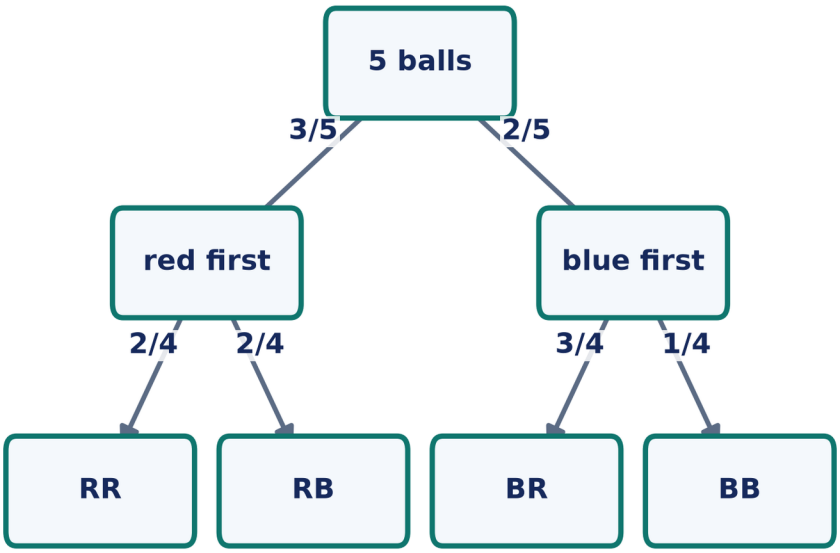
The variance formula here uses divisor N for a complete data set / probability distribution. The unbiased sample estimator with divisor $N - 1$, confidence intervals, and hypothesis testing are outside this syllabus-oriented set.

LESSON 09 / 09

A probability tree multiplies along a path

A bag has three red and two blue balls. Draw two without replacement. Along a path, multiply conditional probabilities; across disjoint paths with the same desired property, add.

Two draws without replacement

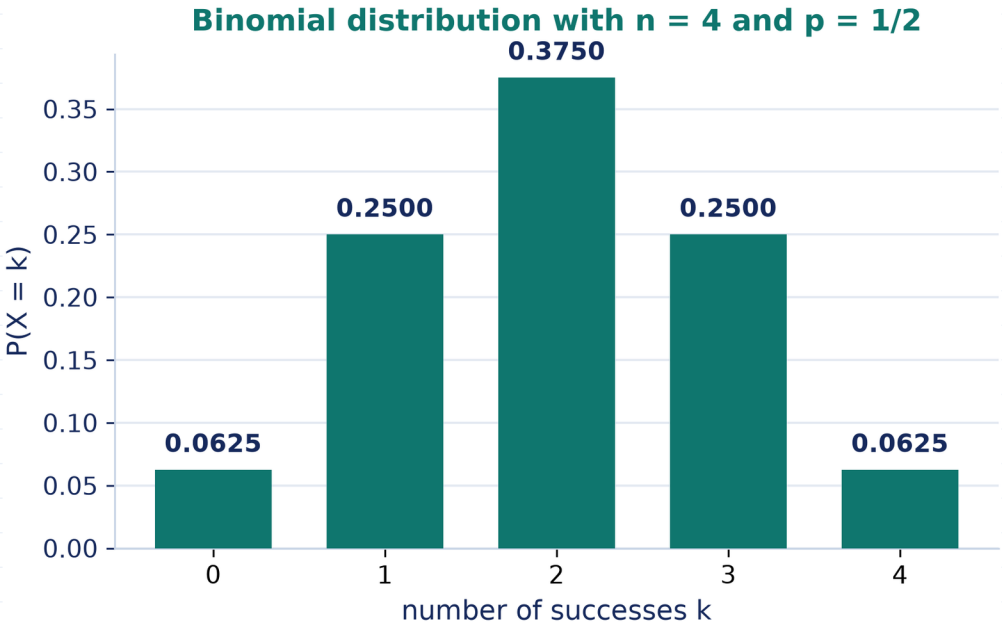


READ THE FIGURE
The second probabilities use four remaining balls. P(same colour) = (3/5)(2/4) + (2/5)(1/4) = 2/5. All four path probabilities sum to 1.

If the first ball is replaced, the second-stage denominators return to 5 and the draws become independent. The sampling rule changes the tree, so it must be read before any probability is assigned.

A distribution lists every value and its probability

For $X \sim \text{Binomial}(4, 1/2)$, $P(X = k) = \binom{4}{k}/16$. The probabilities sum to 1, the mean is $np = 2$, and the variance is $np(1-p) = 1$.



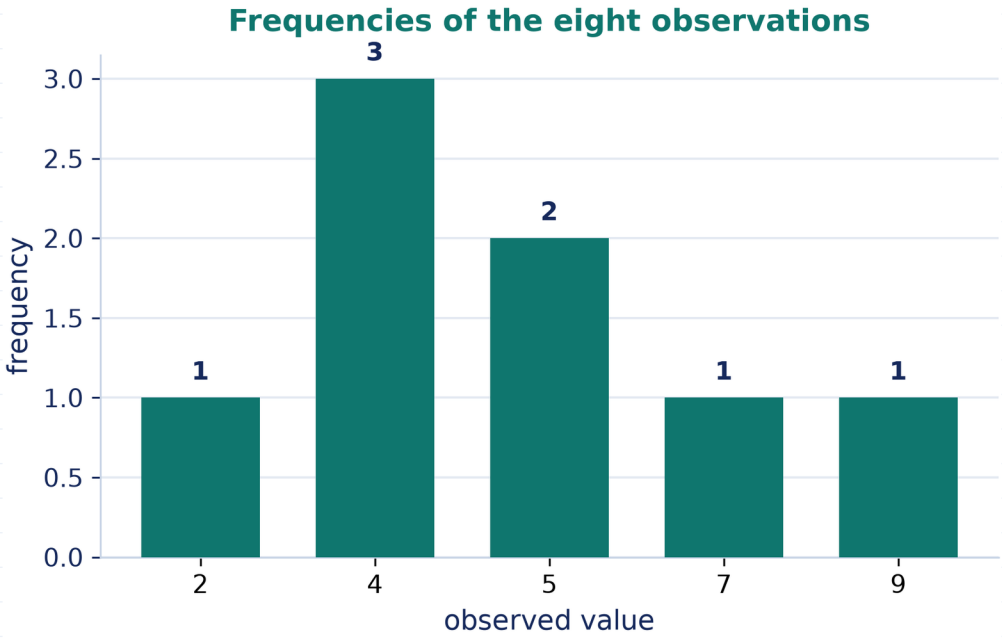
READ THE FIGURE

Symmetry about $k = 2$ reflects $p = 1/2$. The probabilities are $1/16$, $4/16$, $6/16$, $4/16$, $1/16$ and sum to 1.

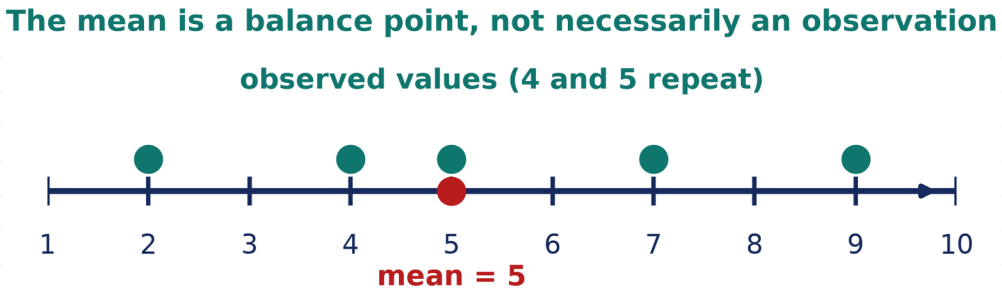
The binomial model requires a fixed number of independent trials, the same success probability on each trial, and exactly two outcomes per trial. Without replacement from a small population, these conditions fail and a hypergeometric model is usually appropriate.

Mean locates; variance measures squared spread

For the data $2, 4, 4, 4, 5, 5, 7, 9$, the mean is 5 and the population variance is 4. Deviations from the mean sum to zero, but squared deviations do not cancel.



READ THE FIGURE
The total frequency is 8 and the weighted sum is 40, so the mean is 5. The squared-deviation total is 32, giving population variance 32/8 = 4.

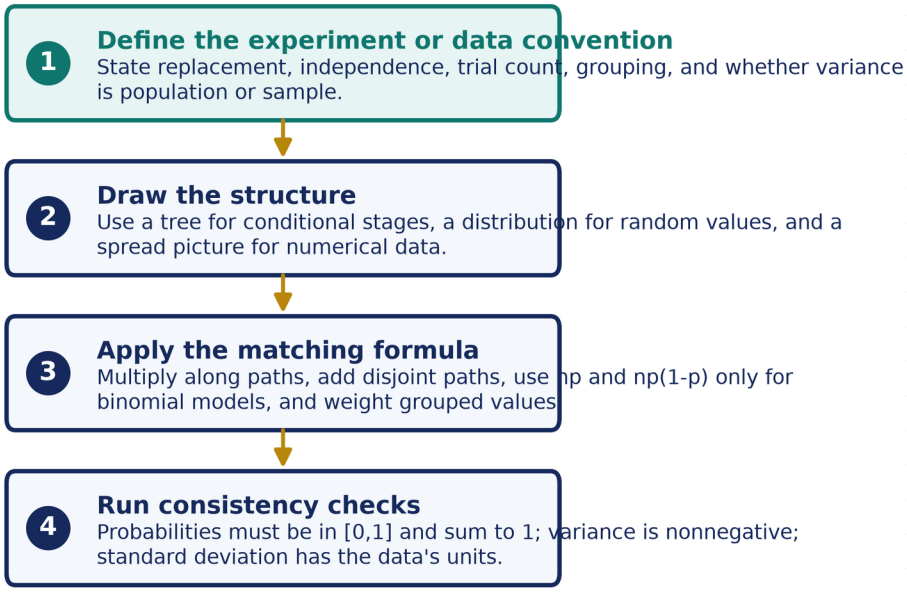


READ THE FIGURE
Positive and negative deviations from 5 cancel. Variance instead averages their squares; standard deviation returns that squared-spread measure to the original units.

For a sample variance with denominator $n - 1$, state that convention explicitly. Do not mix population and sample formulas in one calculation.

One workflow for probability and statistics

Model, calculate, then check the units and total



READ THE FIGURE

A numerical answer is credible only when its model assumptions and units are also correct. The picture makes those assumptions easier to audit.

KEY RESULT

Two checks that catch many errors

For any discrete distribution, recompute $\sum p_i = 1$ and $\sum x_i p_i$ for the mean. For any data set, verify that shifting every value by a constant leaves variance unchanged but shifts the mean by the same constant.

Geometric Probability, Ballot, Birthday

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

LESSON 01 / 03

Continuous sample spaces

Uniformly chosen points in a region: probability is favourable area / total area, provided the joint distribution is uniform in the stated coordinates. Coordinate choice and sampling procedure matter.

WORKED EXAMPLE

Broken stick

Two independent uniform break points on a unit stick make three pieces. Triangle condition is that no piece is at least $1/2$. The region of valid break-point pairs has one quarter the total area, so probability $1/4$.

Bertrand ballot theorem

If $a > b$ votes are randomly ordered, the probability A is strictly ahead after every vote is $(a-b)/(a+b)$. Reflection of paths at their first diagonal contact counts the bad orderings. Strictly ahead excludes ties after the first vote.

Birthday collision

With d equally likely birthdays, independent and ignoring leap day, probability of no match among $n \le d$ people is $\prod_{j=0}^{n-1} (d-j)/d$; collision probability is one minus this. For $d = 365, n = 23$, about 0.5073.

COMMON TRAP

Model assumptions matter

Uniform birthdays, independence, and a 365-day year are approximations. The answer is conditional on this idealised model, not a statement about real populations.

LESSON 02 / 03

Build state equations

For a process with finite memory, let p_s be the desired probability from state s. Condition on the next move: $p_s = \sum_t P(s \to t) p_t$, using boundary values for winning/losing states. For expectations, add one unit of cost per step: $E_s = 1 + \sum_t P(s \to t) E_t$.

WORKED EXAMPLE

Alternating first-head game

Fair coin; players alternate, first head wins. Let p be first player's win chance. She wins now with $1/2$ or both tosses fail with $1/4$ and state resets: $p = 1/2 + p/4$, hence $p = 2/3$.

Coupon collector

For n equally likely types, when j distinct types are collected the next-new-type chance is $(n-j)/n$. Summing geometric stage means gives

$$E[T] = n\left(1 + \tfrac{1}{2} + \cdots + \tfrac{1}{n}\right) = nH_n.$$

Memorylessness and runs

Geometric waiting time: $P(X > m+n \mid X > m) = P(X > n)$. There is no "due" effect. Expected tosses to obtain r consecutive heads obey a run-state recurrence; for a fair coin it is $2^{r+1} - 2$.

LESSON 03 / 03

Gambler's ruin

Fortune starts at i units, absorbing barriers $0, N$, and each unit bet wins with probability p, loses with $q = 1 - p$. Success probability solves $u_i = pu_{i+1} + qu_{i-1}$, $u_0 = 0, u_N = 1$:

$$u_i = \frac{1 - (q/p)^i}{1 - (q/p)^N} \quad (p \neq q), \qquad u_i = i/N \quad (p = q = 1/2).$$

INTUITION

Fair fortune is a martingale

Expected final fortune equals initial fortune i; final fortune is N with probability u_i and zero otherwise, so $Nu_i = i$.

Derangements by inclusion–exclusion

For n objects assigned uniformly to n positions, probability none fixed is

$$\sum_{k=0}^{n} \frac{(-1)^k}{k!},$$

so $D_n = n! \sum_{k=0}^{n} (-1)^k/k!$ and it tends to $1/e$. At $n = 5$, $D_5 = 44$ and probability $11/30$.

WORKED EXAMPLE

Biased walk, exact value

Take $p = 3/5$, $q/p = 2/3$, barriers 0 and 10, start $i = 5$. Then $u_5 = [1 - (2/3)^5]/[1 - (2/3)^{10}] \approx 0.88364$. The edge makes success more likely than 1/2; the boundary conditions still matter.

COMMON TRAP

Do not overgeneralize optional stopping

The elementary fair-ruin argument is safe for finite barriers and the process stopped at absorption. General martingale stopping claims require hypotheses; do not invoke an unrestricted optional-stopping theorem.

Derangement proof and a small check

Let A_i be the event that object i is fixed. Inclusion–exclusion gives $D_n = \sum_{j=0}^{n} (-1)^j \binom{n}{j} (n-j)! = n! \sum_{j=0}^{n} \frac{(-1)^j}{j!}$. For $n = 3$, this is $6 - 9 + 2 = 2$; probability $1/3$. For $n = 5$, $D_5 = 44$, probability $11/30$. The limit $1/e$ is approached with alternating partial-sum error bounded by the next reciprocal factorial.

Why averaging helps

For a random variable X, $E[X] = \sum_i E[I_i]$ when $X = \sum_i I_i$, regardless of dependence. This linearity powers coupon-collector stage sums and indicator proofs of expected collision counts. Variance does not share this unrestricted additivity.

What these extensions teach

The same ideas recur: complement for rare collisions, inclusion–exclusion for forbidden fixed points, state recursions for sequential games, and expectation decomposed into stages. These models add methods; they are not explicit JEE syllabus promises.